

# Session Objectives

1. Review the key embryological events of the nervous system and apply them to congenital neurological defects
2. Describe and discuss the most common **Neural tube defects** (e.g. spina bifida)
3. Describe and discuss the most **common forebrain (e.g.; holoprosencephaly, lissencephaly) and posterior fossa (e.g.; Chiari and Dandy Walker) anomalies**
4. Describe and discuss **syringomyelia and hydrocephalus**
5. Describe **pre and perinatal brain injuries**

# Topics today

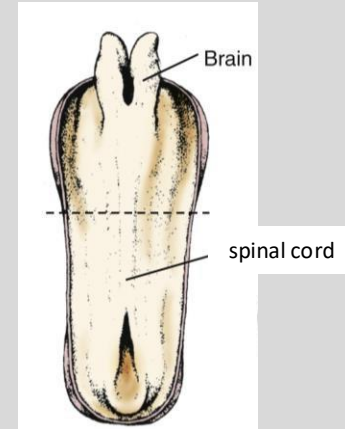
- **Neuroembryology review**
  - Neural tube defects**
    - Spina bifida
    - Myelomenigocele
    - Meningocele
    - Encephalocele
    - Anencephaly
  - Disorders of neuronal migration**
    - Lissencephaly
    - Schizencephaly
    - Neuronal heterotopias
    - Polymicrogyrias
- **Agenesis/dysgenesis disorders**
  - Corpus callosum agenesis
  - Holoprosencephaly
  - Microcephaly
  - Megalencephaly
- **Posterior fossa anomalies**
  - Chiari malformations
  - Dandy-Walker
- **Involving CSF**
  - Syringomyelia
  - Hydrocephalus
- **Cerebral palsy**

# Neural tube defects

- Failure of neural tube/ neuropores to close around **week 3-4** → persistent connection between amniotic cavity and spinal canal
- Maternal serum/amniotic **alpha-fetoprotein** ↑ (except **spina bifida occulta** → **normal** AFP), and **amniotic acetylcholinesterase** ↑ (week 16-18 sampling)
  - Many risk factors: Drugs (**valproic acid, carbamazepine, TMP-SMX, phenytoin, phenobarbital, primidone**), **maternal malnutrition, alcoholism, low folate levels, maternal obesity/diabetes**, and **genetic** determinants (ex. mutations in folate-responsive or folate- dependent enzyme pathways)
  - **Folate** insufficiency is associated with **NTDs**: Current recommendations, potentially become pregnant, consume at least 0.4 mg/day of folic acid
  - **Maternal diabetes** significantly increases risk of NTDs
- Comprise most CNS congenital anomalies and include (covered here) **spina bifida, meningocele, myelomeningocele, encephalocele, and anencephaly**

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Youmans and Winn  
Neurological Surgery 8th  
Ed.

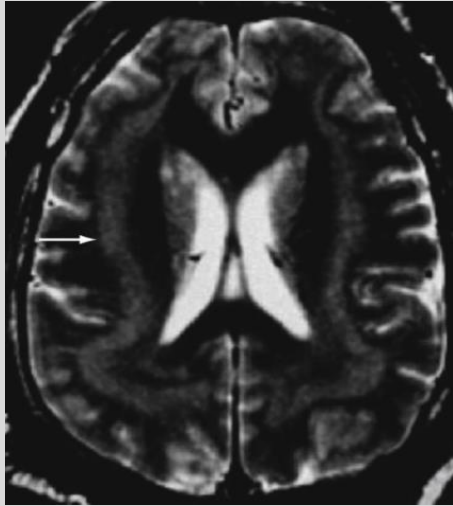


# Neuronal heterotopias

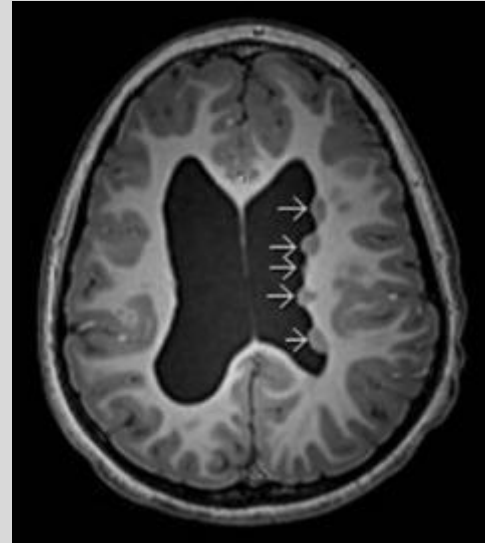
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- Characterized by **altered gyral patterning/organization** or presence of **ectopic neurons in subcortical white matter or periventricular region**
- Several subtypes, main ones shown
- **Intractable seizures** are a common feature
- Several genes have been identified that are a cause of these conditions



Subcortical band heterotopia: Thin layer of gray-matter inside the white matter - **neurons did not migrate to surface**



- Periventricular nodular heterotopia: **Neurons did not migrate superficially from the edges of the neural tube**

# Agensis/dysgenesis disorders

- Brain developmental disorders due to **defective formation of the prosencephalon** and inadequate induction of forebrain structures
  - (Prosencephalon is the embryologic structure that develops into the diencephalon [**neocortex**] and telencephalon [**thalamus/hypothalamus**])

# Chiari malformations

- A **collection** of hindbrain **abnormalities** ranging from simple **herniation** of cerebellar tonsils through foramen magnum to complete **agenesis** of cerebellum
- Likely multifactorial
- **Most symptomology** comes from **abnormal CSF dynamics** across cranio-cervical junction - guides treatment

# Presentation of Chiari malformations

- Chiari I
  - Congenital, usually asymptomatic in **children** & manifests in **adults**
  - **Varies** from subclinical to **headaches** to severe **myelopathy & brainstem dysfunction**
  - **Most common** symptom is **occipital headaches**, which get **worse with Valsalva** maneuvers (laughing, coughing, sneezing) - these actions increase ICP because increased intrathoracic pressure decreases blood return from head (internal jugular vein) & transient venous brain congestion causes symptoms
  - Up to 70-80% have some degree of **ophthalmologic, otologic, or cerebellar dysfunction**
  - Associated with spinal cavitations (ex. **Syringomyelia** in a fluid filled cyst called a **syrinx**)
  - Lower **cranial nerve dysfunctions** are common (7-12)
  - Usually managed **conservatively** with **follow-up and imaging**
- Chiari II
  - Herniation through foramen magnum causes noncommunicating hydrocephalus due to aqueduct stenosis & they develop lumbosacral myelomeningocele
  - >95% of patients with **myelomeningocele** have CMII (**leading cause of death due to hydrocephalus**)
  - ~ 1/3<sup>rd</sup> develop **brainstem symptoms by age 5**, and > 1/3<sup>rd</sup> of those **die, usually of respiratory failure**
  - Usually presents as a neurosurgical **emergency** – goal = **relieve pressure from brainstem** and allow **CSF flow** to spinal canal

# Other Chiari malformations

- **Chiari III** is **rare** and **most severe**, profound herniation and brainstem dysfunction, brain tissue protrudes through skull, **seldom compatible with life**
- **Chiari IV** includes **cerebellar hypoplasia + herniation**
- (There is also Chiari 0, syringomyelia with no tonsil herniation; and Chiari 1.5 is in between severity of type I and II classifications)

# Hydrocephalus

- **Most common** condition treated by **pediatric neurosurgeons**
- Hydrocephalus is not a specific disease - a **diverse group** of conditions that result from **impaired circulation** and/or absorption of **CSF**
- **Congenital** causes include:
  - Aqueduct stenosis
  - Dandy-Walker
  - Holoprosencephaly
  - Myelomeningocele
  - Etc.
- **Acquired** Causes include:
  - Intraventricular hemorrhage
  - Brain tumors
  - Post-surgical (scarring/adhesions)
  - Post-traumatic